

FULL TRAINING/TEST CURVES FOR RNN ON SEQUENTIAL-MNIST

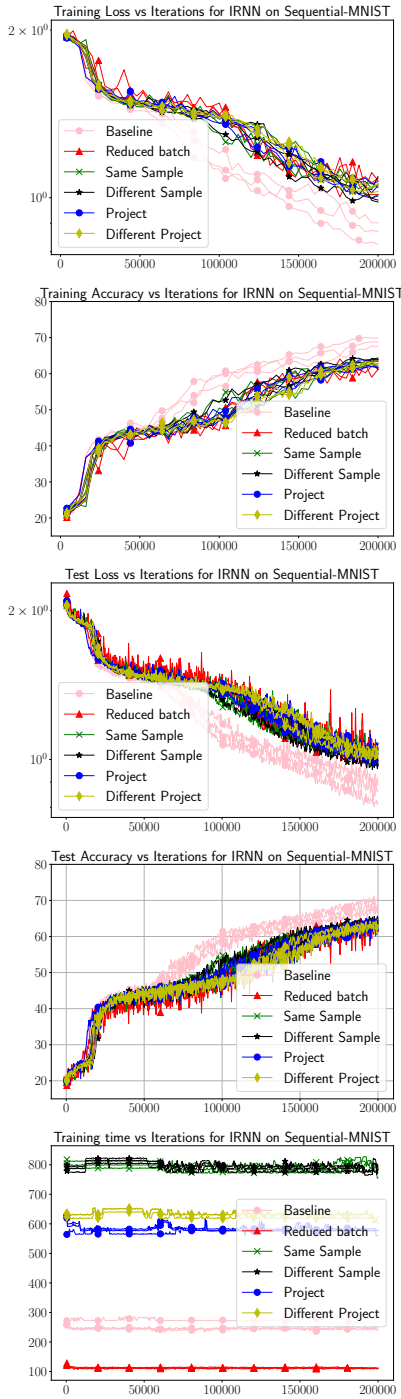


Figure 9: Full train/test curves and runtime per 400 iterations. We also include results for random projections with shared and different random matrices for each mini-batch element.

SWEEP OF FRACTIONS SAMPLED FOR MNIST AND CIFAR-10

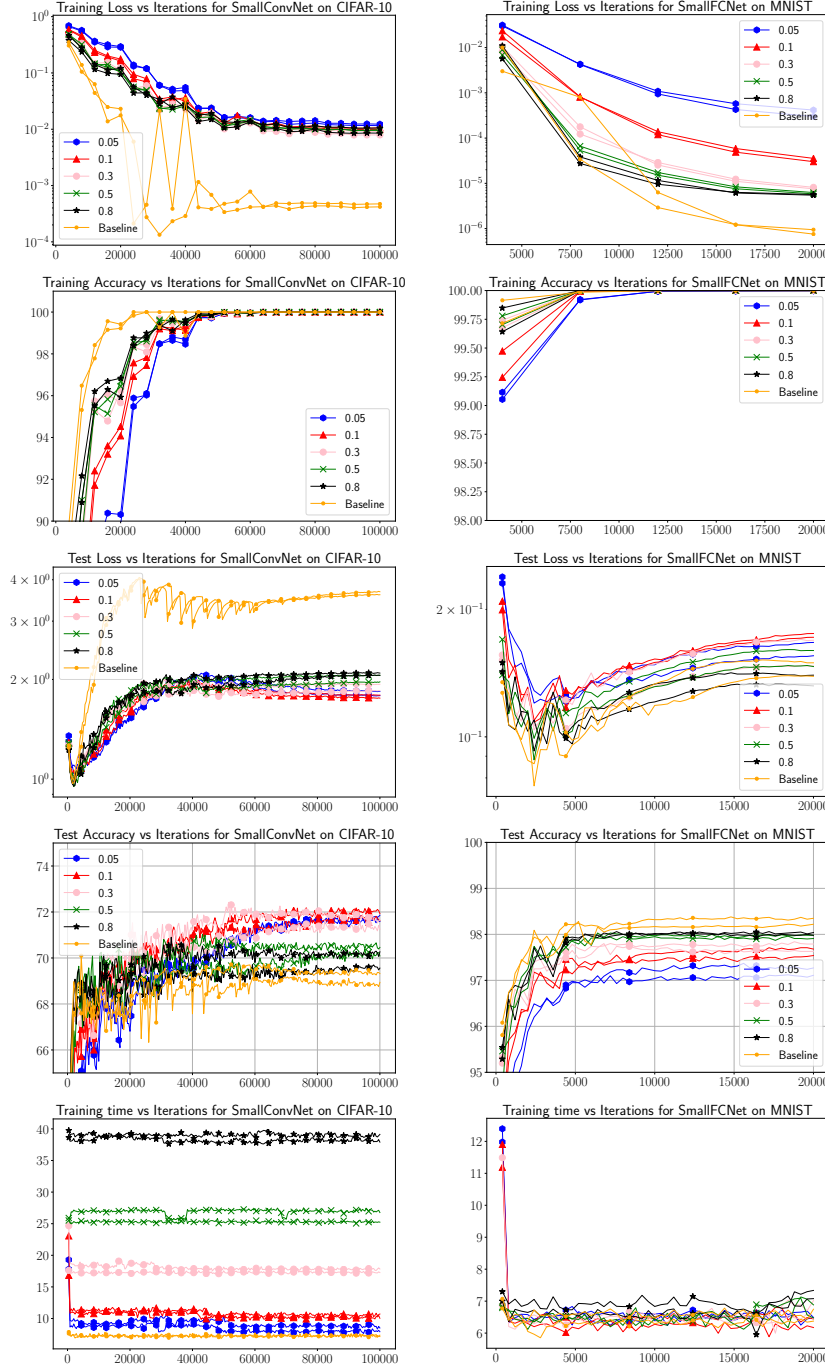


Figure 10: Full train/test curves and runtime per iteration for various fractions for the “different sample” experiment. Note that the reason 0.8 does not quite converge to baseline in the training curve is because we sample with replacement. This is an implementation detail; our method could be modified to sample without replacement, and at fraction 1.0 would be equivalent to baseline. The weight decay and initial learning rate for the RAD experiments above are all the same as the ones tuned for 0.1 fraction “different sample” experiment. The baseline experiments are tuned for baseline.

APPENDIX B: REACTION-DIFFUSION PDE

The reaction-diffusion equation is a linear parabolic partial differential equation. In fission reactor analysis, it is called the one-group diffusion equation or one-speed diffusion equation, shown below.

$$\frac{\partial \phi}{\partial t} = D \nabla^2 \phi + C \phi + S$$

Here ϕ represents the neutron flux, D is a diffusion coefficient, and $C\phi$ and S are source terms related to the local production or removal of neutron flux. In this paper, we solve the one-speed diffusion equation in two spatial dimensions on the unit square with the condition that $\phi = 0$ on the boundary. We assume that D is constant equal to $1/4$, $C(x, y, t, \theta)$ is a function of control parameters θ described below, and S is zero. We discretize ϕ on a regular grid in space and time, which motivates the notation $\phi \rightarrow \phi_t$. The grid spacing is $\Delta x = 1/32$ and the timestep is $\Delta t = 1/4096$. We simulate from $t = 0$ to $t = 10$. We use the explicit forward-time, centered-space (FTCS) method to timestep ϕ . The timestep is chosen to satisfy the stability criterion, $D\Delta t/(\Delta x)^2 \leq \frac{1}{4}$. In matrix notation, the FTCS update rule can be written $\phi_{t+1} = M\phi_t + \Delta t C_t \odot \phi_t$, in index notation it can be written as follows:

$$\phi_{t+1}^{i,j} = \phi_t^{i,j} + \frac{D\Delta t}{(\Delta x)^2} \left(\phi_t^{i+1,j} + \phi_t^{i-1,j} + \phi_t^{i,j+1} + \phi_t^{i,j-1} - 4\phi_t^{i,j} \right) + \Delta t C_t^{i,j} \phi_t^{i,j}$$

The term $C\phi$ in the one-speed diffusion equation relates to the local production or removal of neutrons due to nuclear interactions. In a real fission reactor, C is a complicated function of the material properties of the reactor and the heights of the control rods. We make the simplifying assumption that C can be described by a 7-term Fourier series in x and t , written below. Physically, this is equivalent to the assumption that the material properties of the reactor are constant in space and time, and the heights of the control rods are sinusoidally varied in x and t . ϕ_0 is initialized so that the reactor begins in a stable state, the other parameters are initialized from a uniform distribution.

$$C(x, y, t, \theta) = \theta_0 + \theta_1 \sin(\pi t) + \theta_2 \cos(\pi t) + \theta_3 \sin(2\pi x) \sin(\pi t) + \theta_4 \sin(2\pi x) \cos(\pi t) + \theta_5 \cos(2\pi x) \sin(\pi t) + \theta_6 \cos(2\pi x) \cos(\pi t)$$

The details of the stochastic gradient estimate and optimization are described in the main text. The Adam optimizer is used. Each experiment of 800 optimization iterations runs in about 4 hours on a GPU.